

SubsiGuard

Mine subsidence monitoring & early warning

Problem Statement ID: SIH26025

Problem Statement Title:- Development of an AI-enabled Low Cost Real Time Mine Subsidence Monitoring, Prediction and Early Warning System for Underground Coal Mines in India.

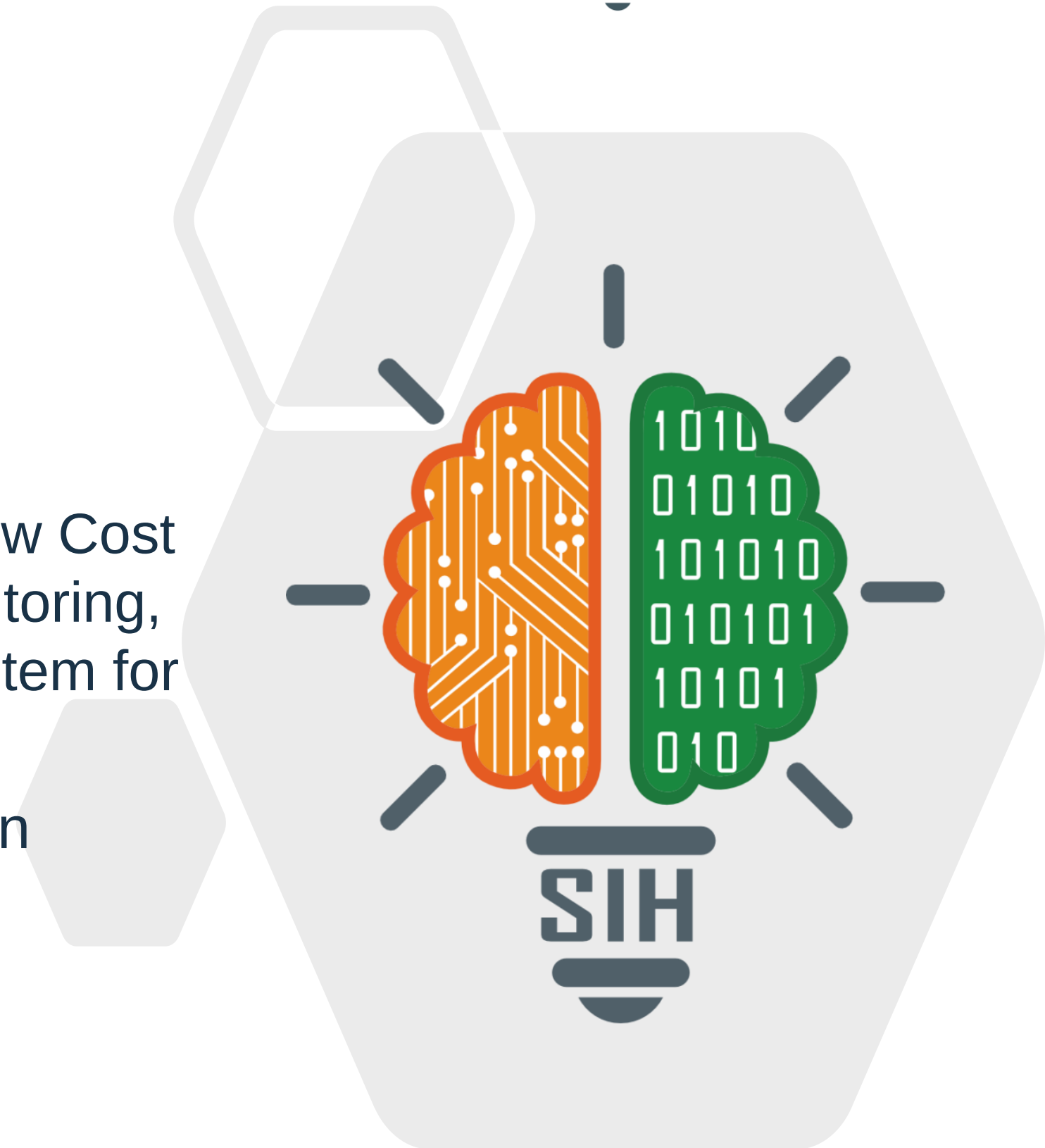
Theme: Smart Automation

PS Category: Hardware

Team Name:

MineNova6

Team ID: [To be assigned]



❖ Proposed Solution (Describe your Idea / Solution / Prototype)

EXISTING METHODS & MONITORING GAP

Satellite InSAR: Radar works through clouds. Periodic revisit and processing cannot provide continuous local telemetry.

Optical ground surveys: Periodic field observations leave gaps between measurements and may expose surveyors to unstable ground.

Microseismic monitoring: Tracks seismic activity. Direct surface tilt and displacement sensing provides complementary deformation evidence.

Borehole extensometers: Measure movement at instrumented locations. Drilling and installation needs vary by site.

SUBSIGUARD: LOCALIZED SURFACE MESH

Continuous sensing: Surface nodes above underground coal mine panels monitor tilt, vibration, relative displacement and cracks.

Hybrid analysis: 100 Hz local sampling and filtering/features feed 1 Hz feature telemetry, Random Forest risk classes and a physics-based trend estimator.

Low-cost network: Custom multi-hop LoRa P2P mesh in India's 865–868 MHz band, with local buffering and solar-assisted power.

Independent fail-safe: ESP32 thresholds trigger local indication. Gateway/local relay siren works without cloud. SMS/app alerts need gateway connectivity.

THE CRITICAL GAP IDENTIFIED

Continuous real-time surface deformation monitoring above underground mine panels, with localized wireless sensing and offline early warning.

❖ Technologies to be used & Process Workflow Architecture

1. SURFACE SENSOR PROTOTYPE

ESP32 node: MPU-6050 tilt/vibration and wired linear displacement sensor.
Relative movement: Anchored node-pair baselines or cracks. LoRa RSSI is not a displacement measure.
Calibration: Prototype tilt accuracy and displacement resolution require validation.

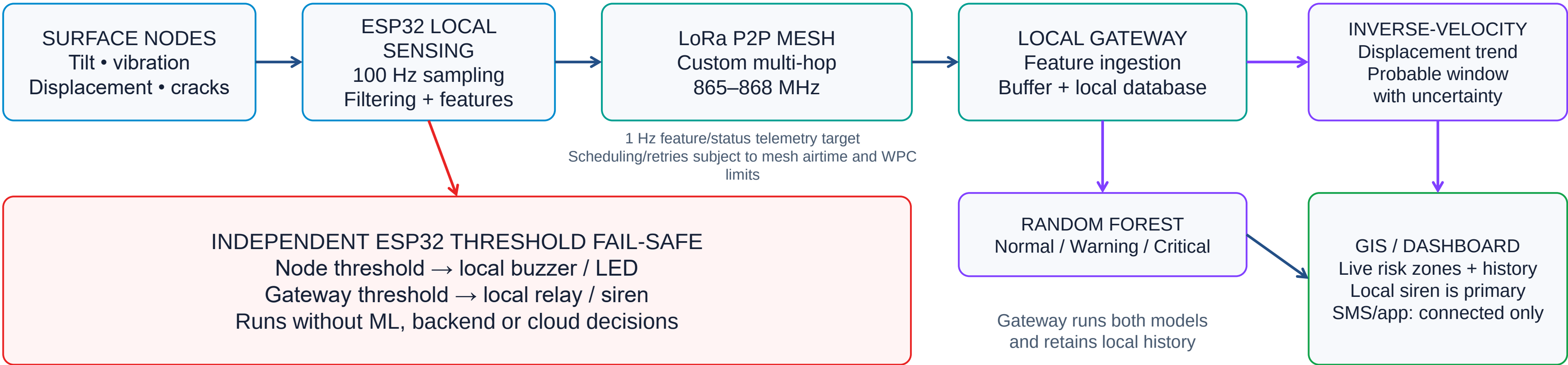
2. HYBRID RISK ANALYSIS

Random Forest: Normal / Warning / Critical from deformation features.
Inverse velocity: Probable failure trend/window only with a sustained acceleration trend.
Prototype ML data: Controlled/simulated scenarios. Real mine data and field calibration required.

3. SPATIAL MODEL & LOCAL SAFETY

Knothe model: Physics-informed spatial/calibration layer. Configurable geometry and strata.
Offline GIS: Local maps and data storage. Sync when connectivity returns.
ESP32 thresholds: Independent node/gateway alarm path bypasses ML, backend and cloud.

SURFACE MONITORING WORKFLOW



Node indication remains available during mesh/gateway outage. Gateway siren needs a working local link.

❖ Analysis of Feasibility, Potential Challenges & Overcoming Engineering Strategies

CHALLENGES ABOVE UNDERGROUND PANELS

Harsh surface conditions: Coal dust, monsoon rain, temperature variation, vegetation and machinery traffic affect nodes and anchors.

Surface radio obstructions: Terrain and buildings can interrupt links. Remote locations may lack cellular coverage.

False alarms and missed events: Traffic vibration, sensor drift and abrupt ground failure challenge classification and trend estimation.

Power and maintenance: Solar availability, relay duty and mesh forwarding affect energy use. Anchors and sensors need inspection.

ENGINEERING STRATEGIES & VALIDATION

Weather protection: Gasketed enclosure and hydrophobic vent. Dust/water ingress performance requires enclosure testing.

Resilient mesh: Node IDs, forwarding, acknowledgements and retries. Local ring buffers reduce data loss during outages.

Alarm robustness: Filtering, multi-sensor agreement and configurable persistence reduce nuisance alarms. Independent thresholds and watchdogs remain active.

Solar-assisted low power: Duty scheduling and replaceable battery. Autonomy requires measured sensing, forwarding and alarm power validation.

ESTIMATED PROTOTYPE NODE COST

₹2,100 / node: Planning estimate, pending itemized BOM validation. Gateway, installation and deployment costs assessed separately.

SCALABLE STUDENT PROTOTYPE

Modular deployment: Readily available ESP32/sensor modules. Site-specific anchors and baseline surveys. Validate airtime before adding nodes.

FIELD SERVICEABILITY

Replaceable components: Modular sensors, battery and enclosure. Scheduled calibration and maintenance. Service life requires field testing.

❖ Expected Benefits, Prototype Validation & Monitoring Support

1. COMMUNITY & WORKER SAFETY

Earlier indication: Detect developing surface movement above underground coal panels to support operator response.

Offline warning: Local siren is the primary alert. Connected gateways can send secondary SMS/app notifications.

Safer observation: Reduce routine exposure of survey crews. Ground surveys and site response plans remain necessary.

Honest warning limits: No guaranteed evacuation lead time. Abrupt failures may lack a useful precursor trend.

2. INFRASTRUCTURE & AFFORDABILITY

Risk visualization: GIS overlays measured deformation and interpreted risk zones near roads, railways and utilities.

Operational decisions: Trend histories help prioritize inspection and mitigation by qualified mine personnel.

Low-cost coverage: Estimated prototype node BOM supports a dense pilot network using accessible hardware.

Comparable budgeting: Compare complete deployments with equivalent coverage and performance after validation.

3. LAND & ENVIRONMENTAL MONITORING

Surface crack tracking: Help identify locations requiring investigation and approved remedial action.

Sensitive land uses: Monitor movement near agricultural land, forests and water infrastructure above mine panels.

Monitoring evidence: Support environmental assessment. Sensors alone cannot prevent fires, aquifer damage or subsidence.

4. REPORTING & DEPLOYMENT READINESS

Regulatory support: Designed to support monitoring/reporting aligned with relevant DGMS/CMR requirements.

CMR 2017: Reg. 111: Development work. Reg. 112: Depillaring operations.

Prototype evidence: ML uses controlled/simulated scenarios. Field calibration and real mine data required before deployment.

Reviewable records: Timestamped telemetry, alarms and calibration logs support review, not certified compliance.

❖ Verified Official Sources & Geotechnical Literature

[\[1\] DGMS / Ministry of Labour & Employment \(2017\)](#) The Coal Mines Regulations, 2017. Reg. 111: Development work. Reg. 112: Depillaring operations. Official Gazette text.

[\[2\] Ministry of Coal \(25 June 2025\)](#) Cabinet approval of the Revised Jharia Master Plan for fire, subsidence and rehabilitation. Release ID 2139495.

[\[3\] Misa, R. \(2023\)](#) Knothe's theory parameters – computational models and examples of practical applications. Mineral Resources Management 39(4), 157–180. DOI: 10.24425/gsm.2023.148164.

[\[4\] Saito, M. \(1965\)](#) Forecasting the Time of Occurrence of a Slope Failure. Proceedings, 6th ICSMFE, Montréal, Vol. 2, pp. 537–541. ISSMGE archive.

[\[5\] Fukuzono, T. \(1985\)](#) A Method to Predict the Time of Slope Failure Caused by Rainfall Using the Inverse Number of Velocity of Surface Displacement. Landslides 22(2), 8–13. DOI: 10.3313/jls1964.22.2_8.

[\[6\] Department of Telecommunications / WPC \(2021\)](#) Use of Low Power Equipment in the Frequency Band 865–868 MHz for Short Range Devices (Exemption from Licence) Rules, 2021. G.S.R. 853(E), 10 December 2021.

[\[7\] ISRO \(2023\)](#) PSLV-C56 / DS-SAR Mission. Official mission description confirms all-weather, day-and-night SAR coverage.